Cambridge Assessment International Education

Cambridge International AS & A Level

MATHEMATICS **9709/43**

Paper 4 Mechanics **October/November 2022**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathematics Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

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Question	Answer	Marks	Guidance
1(a)	$u = 30$	**B1**	From $v = u + at$ or equivalent.
		1	
1(b)	$0 = 30^2 - 2\times10\times s$	**M1**	Use of *suvat* formulae.
	Greatest height $= 45\,\text{m}$	**A1 FT**	FT $u^2/2g$ with *their u* from **(a)**.
		2	

Question	Answer	Marks	Guidance
2(a)	PE = $5g\times15\times1.8\sin20$	**M1**	Attempt to find PE gain.
	PE = 462 J	**A1**	From 461.727…
		2	
2(b)	WD = $5g\times15\times1.8\sin20 + 40\times15\times1.8$ **or** WD = $(5g\sin20 + 40)\times15\times1.8$	**M1**	Uses WD by pulling force = PE gain + WD against friction or WD = Fs.
	WD = 1540 J	**A1 FT**	From 1541.727… FT '1080 + PE from **(a)**'.
		2	

Question	Answer	Marks	Guidance
3	$T\cos 25 = 40 + R\cos 50$	**M1**	Resolving in any direction e.g. horizontal, vertical, along radius or tangent.
	$R\sin 50 = T\sin 25$	**M1**	Resolving in a second direction.
	Radially: $T\cos 25 = R + 40\cos 50$ Tangentially: $T\sin 25 = 40\sin 50$ Parallel to T: $T = R\cos 25 + 40\cos 25$ Perpendicular to T: $R\sin 25 = 40\sin 25$ Vertically: $T\cos 25 = 40 + R\cos 50$ Horizontally: $R\sin 50 = T\sin 25$	**A1**	Two correct equations.
	Solving equation(s) to find either T or R	**M1**	
	$T = 72.5$ N	**A1**	From 72.504….
	$R = 40$ N	**A1**	
		6	

Question	Answer	Marks	Guidance
4(a)	$52 = u(2) + 0.5a(2)^2 - \left(u(1) + 0.5a(1)^2\right)$ **or** $64 = u(4) + 0.5a(4)^2 - (u(3) + 0.5a(3)^2$	**M1**	Use of $s = ut + ½\ at^2$ or equivalent to form equation for 2nd **or** 4th second .
		M1	Second equation in u and a.
	$52 = u + 1.5a$ and $64 = u + 3.5a$	**A1**	Two correct equations in u and a.
	$12 = 2a$ leading to $a = 6$	**M1**	Solves simultaneous equations to find either u or a.
	Initial speed $= 43\ \text{ms}^{-1}$ and acceleration $= 6\ \text{ms}^{-2}$	**A1**	
		5	
4(b)	$s = 43 \times 10 + 0.5 \times 6 \times 10^2$	**M1**	Use of $s = ut + ½\ at^2$ or equivalent.
	Distance $= 730\,\text{m}$	**A1 FT**	FT $10u + 50a$ with u and a from part **(a)**.
		2	

Question	Answer	Marks	Guidance
5(a)	$v = \int 12t + 12 \text{ d}t$, $v_X = 6t^2 + 12t$ **or** $v = \int 24t - 8 \text{ d}t$, $v_Y = 12t^2 - 8t$	**M1**	Uses $v = \int a \text{ d}t$.
	$v_X = 6t^2 + 12t$ $v_Y = 12t^2 - 8t$	**A1**	All correct.
	$6t^2 + 12t = 12t^2 - 8t$ leading to $t = \frac{10}{3}$	**B1**	Solves $v_X = v_Y$.
	$s_X = \int 6t^2 + 12t \text{ d}t$ $s_X = 2t^3 + 6t^2$ $s_Y = \int 12t^2 - 8t \text{ d}t$ $s_Y = 4t^3 - 4t^2$	***M1**	Uses $s = \int v \text{ d}t$.
	$s_X = 2\left(\frac{10}{3}\right)^3 + 6\left(\frac{10}{3}\right)^2$ $s_Y = 4\left(\frac{10}{3}\right)^3 - 4\left(\frac{10}{3}\right)^2$	**DM1**	Evaluates $s\left(\frac{10}{3}\right)$ for each particle.
	$AB = \frac{3800}{27} - \frac{2800}{27} = \frac{1000}{27}$ Distance AB = 37 m	**A1**	
		6	
5(b)	$AB = -2t^3 + 10t^2 = -2 \times 3^3 + 10 \times 3^2$	**M1**	Calculates distance AB.
	$[t = 3]$, AB = 36 m	**A1**	AG
		2	

Question	Answer	Marks	Guidance
6(a)(i)	$P = (650 + 150) \times 24$	**M1**	Use of $P = \text{DF} \times v$.
	19 200 W or 19.2 kW	**A1**	
		2	
6(a)(ii)	$40\,000 = \text{DF} \times 24$	**B1**	Correct use of $P = \text{DF} \times v$.
	$\frac{40000}{24} - 800 = 2250a$	**M1**	Use of Newton's Second Law for the system or for the caravan or for the car.
	$T - 150 = 500a$	**A1**	Two correct equations.
	$a = \left(\frac{40000}{24} - 800\right) \div 2250$ leading to $a = \ldots$	**M1**	Solves for a or for T.
	Acceleration $= 0.385\ \text{ms}^{-2}$ and Tension $= 343\,\text{N}$	**A1**	From $a = \frac{52}{135} = 0.38518\ldots$ and $T = \frac{9250}{27} = 342.59\ldots$
		5	
6(b)	$\text{DF} = 800 + 2250 \times g \times 0.14$	**M1**	Resolving up hill using DF = Total resistances.
	$\frac{31000}{v} = 800 + 2250 \times g \times 0.14$	**M1**	Use of $P = \text{DF} \times v$ to form equation in v.
	$v = 7.85$	**A1**	From $v = \frac{620}{79} = 7.848\ldots$
		3	

Question	Answer	Marks	Guidance
7(a)	$R = 3g\cos\alpha = 3\times10\times0.8$	**B1**	
	$F = 3g\sin\alpha = 3\times10\times0.6$	**M1**	Resolving parallel to plane.
	$\mu = \frac{18}{24} = 0.75$ **or** $\mu = \frac{3g\sin\alpha}{3g\cos\alpha} = \tan\alpha = 0.75$	**A1**	Uses $\mu = \frac{F}{R}$ AG.
		3	
7(b)	$a = g\sin\alpha$ or PE loss $= 1.5gx\sin\alpha$ for *AB* **and** $a = 0$ for *BC* $4.5g\times\sin\alpha - 0.75\times4.5g\cos\alpha = 4.5a$ leading to $a = 0$	**B1**	Accelerations for *AB* and *BC*.
	$v_1^2 = 2\times g\sin\alpha\times x$] **or** $[1.5g\times x\sin\alpha = 0.5\times1.5\times v_1^2$	**M1**	Uses '*suvat*' or PE loss = KE gain for *AB*.
	$v_1^2 = 20x\sin\alpha = 12x$ leading to $v_1 = \sqrt{12x}$	**A1**	
	$1.5\times\sqrt{12x} + 0 = 4.5\times v_2$ leading to $v_2 = \frac{1}{3}\sqrt{12x}$	**M1**	Conservation of momentum.
	$4 = \frac{2}{3}\times\sqrt{12x}$	**M1**	Use of $s = vt$ on *BC* since $a = 0$.
	$x = 3$	**A1**	

Question	Answer	Marks	Guidance
7(b)	**Alternative Method for 7(b)**		
	$a = g\sin\alpha$ or PE loss $= 1.5gx\sin\alpha$ for *AB* **and** $a = 0$ for *BC* $4.5g\times\sin\alpha - 0.75\times 4.5g\cos\alpha = 4.5a$ leading to $a = 0$	**B1**	Accelerations for *AB* and *BC*.
	$4 = 2v_2$ leading to $v_2 = 2$	**M1**	Uses $s = vt$ on *BC* since $a = 0$.
	$1.5\times v_1 + 0 = 4.5\times 2$	**M1**	Conservation of momentum.
	$v_1 = 6$	**A1**	Velocity before collision.
	$6^2 = 2\times g\sin\alpha\times x$ **or** $1.5g\times x\sin\alpha = 0.5\times 1.5\times 6^2$	**M1**	Uses *suvat* or PE loss = KE gain for *AB*.
	$x = 3$	**A1**	
		6	
7(c)	KE = $0.5\times 4.5\times 2^2 = 9\text{J}$	**B1**	KE gain for *AC*.
	PE loss = $15\times(4+3)\times\frac{3}{5} + 30\times 4\times\frac{3}{5} = 135\text{J}$	**M1**	Evaluates PE loss for *AC*.
	Loss of energy = 126 J	**A1**	
		3	